Machine Learning – CDS201

Module 3: Classification Models

Conditional probability

- Let A, B be two events. The conditional probability of A given B (that is, B has occurred), denoted by $P(A|B)$, is defined as

$$P(A|B) = P(A \text{ and } B)/P(B) = P(A \cap B)\ P(B)$$

- The above relation leads to:

$$P(A \text{ and } B) = P(A \cap B) = P(A|B)P(B);$$

$$P(A \text{ and } B) = P(A \cap B) = P(B|A)P(A).$$

Bayes' Theorem

- Let A1, . . . , Ak be a collection of k mutually exclusive and exhaustive events with prior probabilities P(Ai), i = 1, . . . , k.
- Let B be any event with P(B) > 0.
- Then the posterior probability of Ai given that B has occurred is

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)}$$
$$= \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^{k} P(B|A_i)P(A_i)},$$

Analogy to understand Bayes theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- **Rain and Clouds:**
  - A = It will rain today.
  - B = It is cloudy this morning.
- Let's translate the formula:
  - **P(A|B) (Posterior Probability):** The probability that **it will rain** *given that* **it is cloudy**.
  - **P(B|A) (Likelihood):** The probability that **it would be cloudy** *given that* **it rains**. This is easier to measure. We can look at all past rainy days and see how many of them were cloudy. (Probably most of them!)
  - **P(A) (Prior Probability):** The probability that **it will rain**, regardless of any evidence. This is our initial belief. For example, by looking at historical data, we might know that it rains on 10% of days in this city. So, P(A) = 0.10.
  - **P(B) (Marginal Probability):** The probability of **seeing the evidence** (it being cloudy), regardless of whether it rains or not.

The “Naïve” Assumption — A Clever Simplification

- Our goal is to predict a class given a set of features.

- class (A) = Spam, Not Spam
- features (B) = Words in an email

$$P(\text{class}|\text{feature1}, \text{feature2}, \text{feature3}) = \frac{P(\text{feature1},\text{feature2},\text{feature3}|\text{class}) \times P(\text{class})}{P(\text{feature1},\text{feature2},\text{feature3})}$$

- The term P(feature1, feature2, feature3 | class) is incredibly hard to calculate. It means “What’s the probability of seeing this *exact combination* of features, given that the class is ‘Spam’?”
- You would need a gigantic amount of data to have seen every possible combination of features multiple times.

This is where the “NAÏVE” assumption

“Naïve” Bayes

- Instead of calculating the probability of the whole combination, we can just multiply the individual probabilities:

P(feature1, feature2, feature3 | class) ≈ P(feature1 | class) * P(feature2 | class) * P(feature3 | class)

- This is *way* easier to calculate! We just need to find the probability of each individual feature, given the class.

“Naïve” Bayes Classifier

- Naïve Bayes classification rule is:

  - Calculate a “score / probability” for each possible class.
  - Choose the class label with the highest score.

Score(C) = P(C) * P(feature1 | C) * P(feature2 | C) * … * P(featureN | C)

$$\hat{y} = \arg\max_y P(y) \cdot \prod_{i=1}^{n} P(x_i|y)$$

  - Note: *We can ignore the denominator P(features) from the original Bayes’ formula because it’s the same for all classes*

Play-tennis example

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

- Prior Probability

P(p) = 9/14
P(n) = 5/14

- Likelihood for each feature

outlook	
P(sunny|p) = 2/9	**P(sunny|n) = 3/5**
P(overcast|p) = 4/9	**P(overcast|n) = 0**
P(rain|p) = 3/9	**P(rain|n) = 2/5**
temperature	
P(hot|p) = 2/9	**P(hot|n) = 2/5**
P(mild|p) = 4/9	**P(mild|n) = 2/5**
P(cool|p) = 3/9	**P(cool|n) = 1/5**
humidity	
P(high|p) = 3/9	**P(high|n) = 4/5**
P(normal|p) = 6/9	**P(normal|n) = 2/5**
windy	
P(true|p) = 3/9	**P(true|n) = 3/5**
P(false|p) = 6/9	**P(false|n) = 2/5**

Outlook	P	N
sunny	2/9	3/5
overcast	4/9	0
rain	3/9	2/5
Tempreature		
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5

Humidity	P	N
high	3/9	4/5
normal	6/9	1/5
Windy		
true	3/9	3/5
false	6/9	2/5

Classification for the test data

- An unseen sample X = <rain, hot, high, false>

$$P(X|p)\cdot P(p) =$$
$$P(rain|p)\cdot P(hot|p)\cdot P(high|p)\cdot P(false|p)\cdot P(p) =$$
$$3/9\cdot 2/9\cdot 3/9\cdot 6/9\cdot 9/14 = 0.010582$$

$$P(X|n)\cdot P(n) =$$
$$P(rain|n)\cdot P(hot|n)\cdot P(high|n)\cdot P(false|n)\cdot P(n) =$$
$$2/5\cdot 2/5\cdot 4/5\cdot 2/5\cdot 5/14 = 0.018286$$

Sample X is classified in class n (don't play)

Test your knowledge

- What is the primary assumption of the Naive Bayes algorithm?

A. All features are dependent on each other
B. All features are independent of each other
C. All features have equal weight
D. All features have the same mean